RUSHIL BARVE

Hackathon Builder · CS & Computational Mathematics @ UWaterloo



About Me

Olympiad-level problem solver who builds fast, pitches clearly and leads teams under pressure. Eight hackathons from high school in Qatar to Waterloo.

Contact

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LOCATION
Waterloo, Ontario

Skills

LANGUAGES
Python, JavaScript, TypeScript, C, HTML, CSS
FRAMEWORKS
React, Vite, Tailwind CSS, Bootstrap, Zustand
TOOLS
Git, GitHub Actions, IndexedDB (Dexie), Sass, npm, Render

Strengths

- Rapid prototyping & shipping
- Public speaking and pitching
- Team management
- Event logistics
- Marketing and branding

Achievements

- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Six Best Delegate awards across 8+ national and international MUN conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26), Bronze (2023-24, 2024-25)

PROJECTS

CircuitGuard — Python · FastAPI · React · Three.js
Winner, Ampere – AI for Motor Safety · FormulaTech Hacks 2026

- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.
- Built with a team of four at FormulaTech Hacks 2026 (University of Waterloo).

MUN Chair Toolkit — TypeScript · React · Vite · Tailwind · Zustand · Dexie
Local-first web app for Model UN chairs

- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.
- Kept rules in framework-free functions (src/lib/rules.ts) with automated tests; GitHub Actions workflow for checks.
- Stores everything in the browser with IndexedDB, so it works offline with no account or backend.

Track Nector — Python · HTML · JSON · Render
Self-hosted personal finance tracker

- Lightweight tracker for card balances and daily spending built around manual entry, with no bank account linking.
- Chose self-hosting and manual input deliberately to keep financial data private.
- Wrote the account logic in Python (bank_account.py) and added Render deployment config; 11 commits.

Source code available **on request**.

HACKATHONS

FormulaTech Hacks — Winner (Ampere)	2026	RevenueCat Shipaton	2026
OpenCV AI Competition (AWS)	2026	Nebius x NVIDIA Global AI Hackathon	2026
IBM BOB 2.0 Hackathon	2026	DPS MIS Doha Hackathon	2025
Birla School Hackathon — Qatar	2024	CMU Qatar Hackathon	2023

8 hackathons · 5 won · 1 organised

WATERLOO TEAMS

WARG — Waterloo Aerial Robotics Group — 2026
Autonomy team · Embedded Flight Software team

UW Orbital · UW Game Dev — 2026
Student satellite design team · built beginner-level games

Member: Computer Science Club, UW Data Science Club

LEADERSHIP

Director of MUN, DPS-Modern Indian School — May 2025 – Jun 2026

Head of Marketing, TEDxYouth@DPSMIS — Sep – Dec 2025

Head of Operations, DPSDQ MUN — Jun – Oct 2025

Team Leader, Qatar Children's Science Congress — Dec 2024 – Mar 2025